Employee Attendance Management System

System Analysis Documentation

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1. Introduction:

This document aims to analyze and document the Employee Attendance Management System as part of a comprehensive system analysis process to define business requirements and related workflows, The system focuses on automating the process of recording employee attendance and departure, improving data accuracy, and facilitating monitoring of employee compliance with working hours ,It also aims to support management in making effective decisions by providing accurate analytical reports and user-friendly interfaces that enhance daily operations with flexibility and efficiency

1.2 Stackholders:

- **Employee**: Uses the system to check in, check out, view personal attendance performance, and submit justifications for late or early checkouts
- **Admin**: Responsible for managing the system by adding, updating, or deleting employee and department records. Also monitors attendance reports and overall system performance.
- **HR Department**: Reviews attendance data, evaluates justifications submitted by employees, and ensures compliance with attendance policies
- **IT Support / System Analyst**: Maintains the system, handles technical issues, and ensures data integrity and security
- **Management / Department Head:** Uses the generated attendance reports to monitor employee performance and departmental productivity.

1.3 Objective:

The main objective of the Employee Attendance Management System is to simplify the attendance tracking process within the organization, Specifically the system aims to:

- Develop a web-based system to manage employee attendance logs efficiently.
- Automate check-in and check-out processes for employees.
- Enable administrators to manage employee and department records (add, update, delete).
- Provide a performance dashboard for employees and managers.
- Ensure accurate tracking of working hours (7:00 AM – 4:00 PM) and lunch breaks.
- Detect late check-ins and early check-outs automatically.
- Allow employees to justify flagged attendance records when required.
- Generate reports to support decision-making and improve

1.4 Scope:

1.4.1 In-Scope:

The Employee Attendance Management System will include the following features and functionalities:

- **Employee Management (**Add, update, and delete employee information.**)**
- **Department Management** (Add, update, and delete department details.)
- **Attendance Management (**Allow employees to perform daily check-in and check-out actions.)
- **Attendance Validation (**The system automatically flags late check-ins (after 8:30 AM) and early check-outs (before 3:00 PM)).
- **Justification Feature (**Employees can submit justifications for flagged attendance events (optional)).
- **Dashboard and Reporting (**Provide dashboards for employees and managers to monitor attendance performance and working hours).
- **Authentication and Authorization (**Secure login process for both employees and administrators with role-based access)
- **Work Hour Tracking (**Ensure employees complete 40 working hours per week, including a 1-hour lunch break (12:00 PM – 1:00 PM))

1.4.2 Out-Scope:

- Integration with payroll or salary systems.
- Mobile application development (the solution is web-based only).
- External email gateway connection (notifications are simulated)

1.5 System Requirements:

1.5.1 Functional Requirements

FR-1	**The system must allow administrators to:** • **Add, edit, and delete employee records.** • **Add, edit, and delete department records.** • **View attendance dashboards for employees and departments.**
FR-2	The system must allow employees to: • Perform daily **check-in** and **check-out** actions. • View their attendance performance dashboard. • Submit justification for flagged attendance (optional).
FR-3	The system must automatically: • Flag late check-ins (after 8:30 AM). • Flag early check-outs (before 3:00 PM). • Calculate total working hours per week (target 40 hours)

Table 1.1 Functional requirements

1.5.2 Non-Functional Requirements:

NFR-1	**Performance**: The system should respond to user actions (check-in, check-out, report generation) within 2 seconds to ensure smooth performance
NFR-2	**Security:** The system must ensure secure login and protect employee attendance data from unauthorized access. All data should be stored in a secured database with user authentication
NFR-3	**Usability:** The interface should be simple and intuitive, allowing users to easily record attendance and view their records without training
NFR-4	**Reliability:** The system should function correctly under all normal conditions, validate input data, and handle system or network errors gracefully
NFR-5	**Maintainability:** The system should be easy to update or modify by the IT administrator without affecting existing functionalities. Proper documentation should be provided for maintenance

Table 1.2 Non-Functional requirements

2. Wireframes for Solution Screens:

2.1 Login Screen

The first screen allows users (employees and administrators) to access the system. It contains fields for **Username** and **Password**, a **Login** button, and an optional **Remember Me** checkbox, upon successful login, the system proceeds with authentication and authorization processes

System automatically records all logs all user action (login, check in, check out ,etc.)

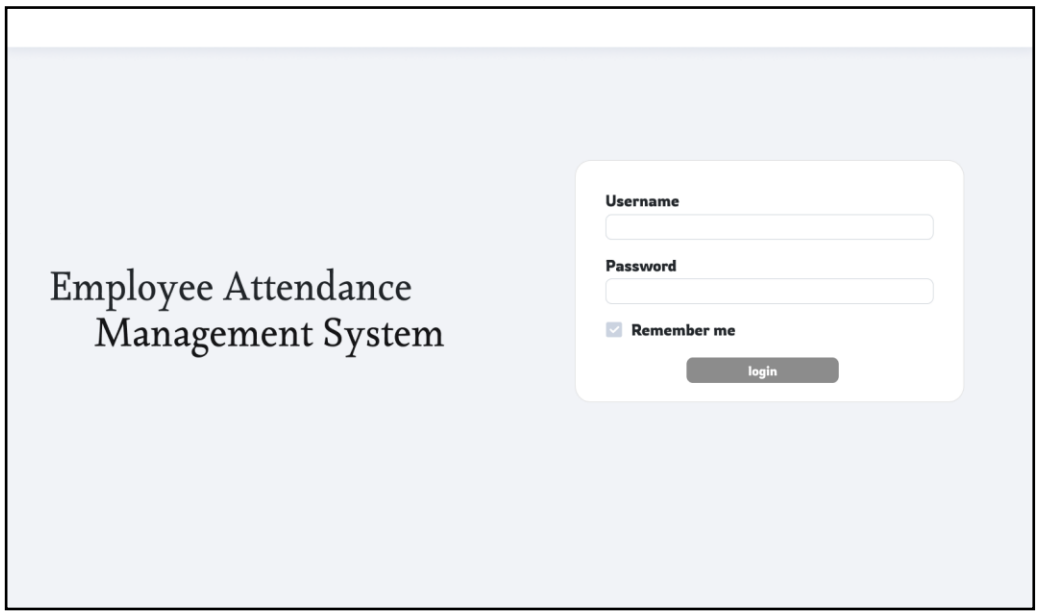


Figure 2.1 Log in Screen

2.2 Authentication

After the user submits login credentials, the system verifies the entered information with the database to confirm identity, If the credentials match, access is granted; otherwise, an error message is displayed.

2.3 Authorization

Once authenticated, the system determines the user's role and redirects them to the appropriate dashboard:

Employee Dashboard – for regular employees to manage attendance.

Admin Dashboard – for administrators to manage employees and departments.

2.4 Employee Screen

The following screens allow employees to manage their daily attendance and monitor their overall performance within the system.

The Attendance Page enables employees to perform check-in and check-out actions for each working day and view their total worked hours per week, If any attendance record is flagged as Late or Early Check-out, the employee can provide justification to explain the reason.

The Performance Dashboard provides a summary of the employee's attendance statistics, including total on-time presence, late arrivals, and early check-outs, it also displays a simple performance chart to visually represent attendance behavior and help the employee track compliance with working hours

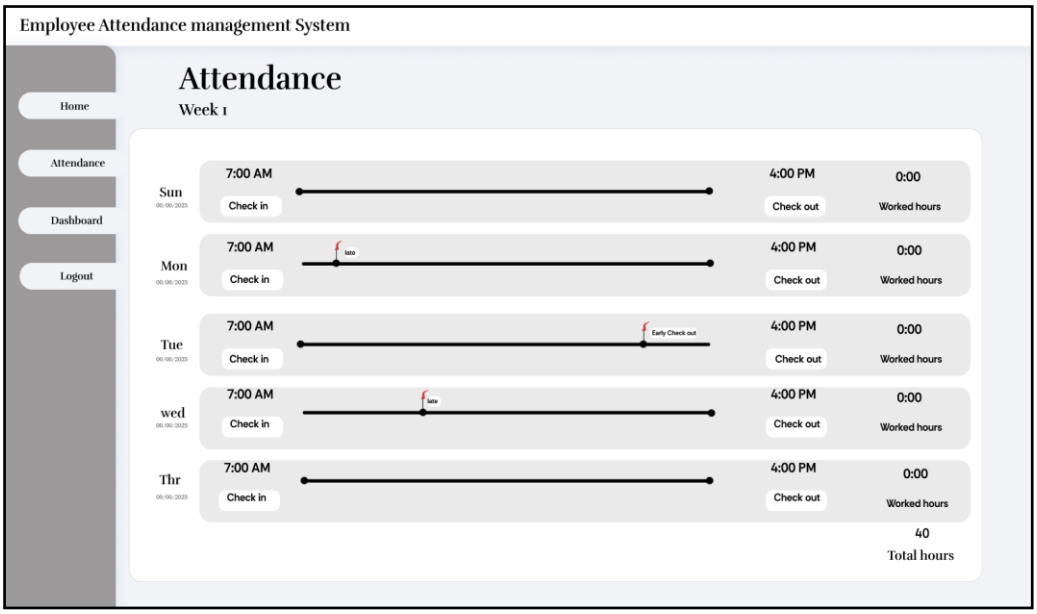


Figure 2.2 Attendance Screen

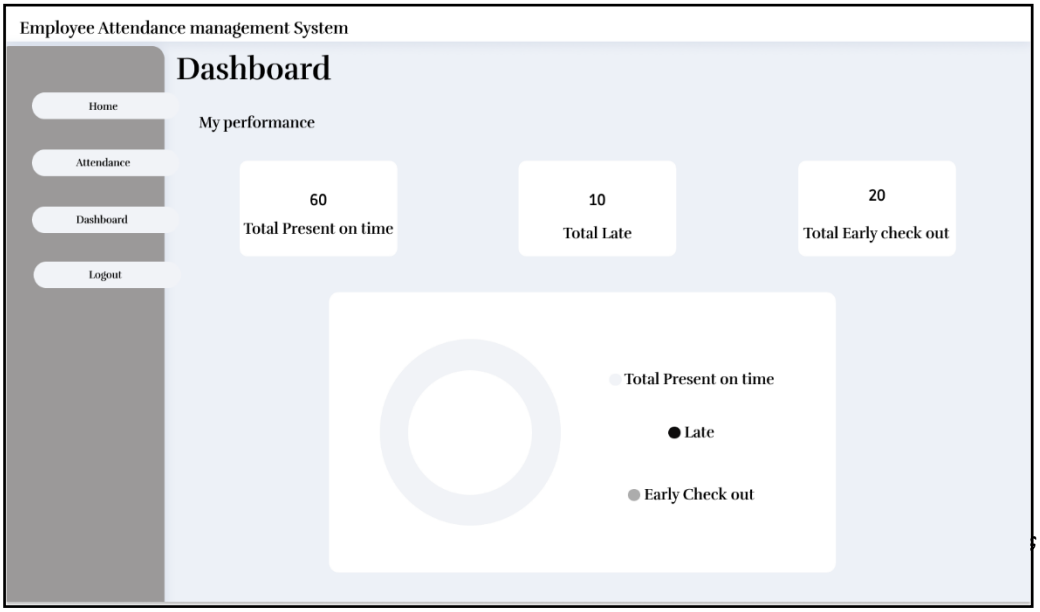


Figure 2.3 Dashboard Screen

2.5 Admin Screen

The following screens represent the administrator's view of the **Employee Attendance Management System**. These pages allow the admin to manage employee and department data, monitor attendance statistics, and ensure accurate organizational reporting.
The admin has access to a summarized dashboard and full control over system records.

1. **Employee Management Page** – provides a list of all registered employees along with their details (Name, ID, Department), the admin can **add**, **update**, or **delete** employee records using the available controls.
2. **Department Management Page** – displays department-level information, including total employees, present, and absent counts, through this screen, the admin can **add**, **modify**, or **remove** department records when needed
3. **Admin Dashboard** – displays an overview of total employees, present employees, and number of departments, it also includes **department-wise attendance analysis**, showing attendance rates for each department to help monitor overall performance

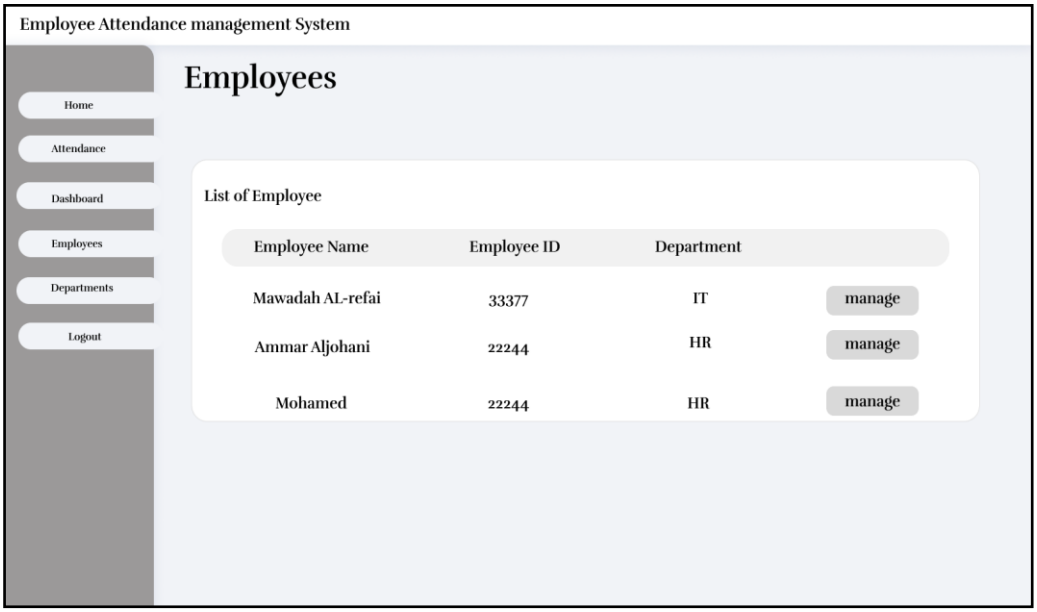


Figure 2.4 Employees List Screen

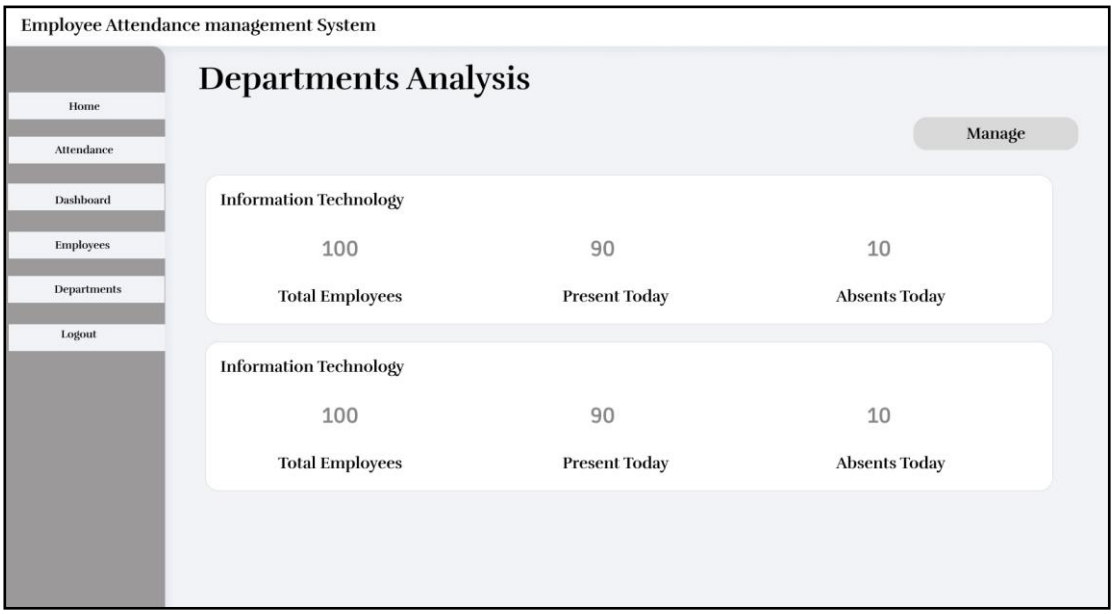


Figure 2.5 Departments Screen



Figure 2.6 Dashboard Admin Screen

3. Databased Model Diagram

The **Entity Relationship Diagram (ERD)** below illustrates the logical structure of the *Employee Attendance Management System* database. It defines the main entities (**Department**, **Employee**, and **Attendance**) and the relationships between them. Each entity includes its primary and foreign keys, ensuring data integrity and efficient access.

- **Department**: Stores information about departments within the organization. Each department has a unique **Department ID (PK)** and a **Department Name**.
- **Employee**: Contains employee details such as **Employee ID (PK)**, **Employee Name**, **Username**, and **Password**. It also includes **Department ID (FK)** to associate each employee with their respective department.
- **Attendance**: Records daily attendance activities, including **Date**, **Check-in**, **Check-out**, and **Justification** (for late or early leave). It references **Employee ID (FK)** to link each attendance record to the corresponding employee.

Relationships:

- One **Department** can have **many Employees**.
- One **Employee** can have **many Attendances records**

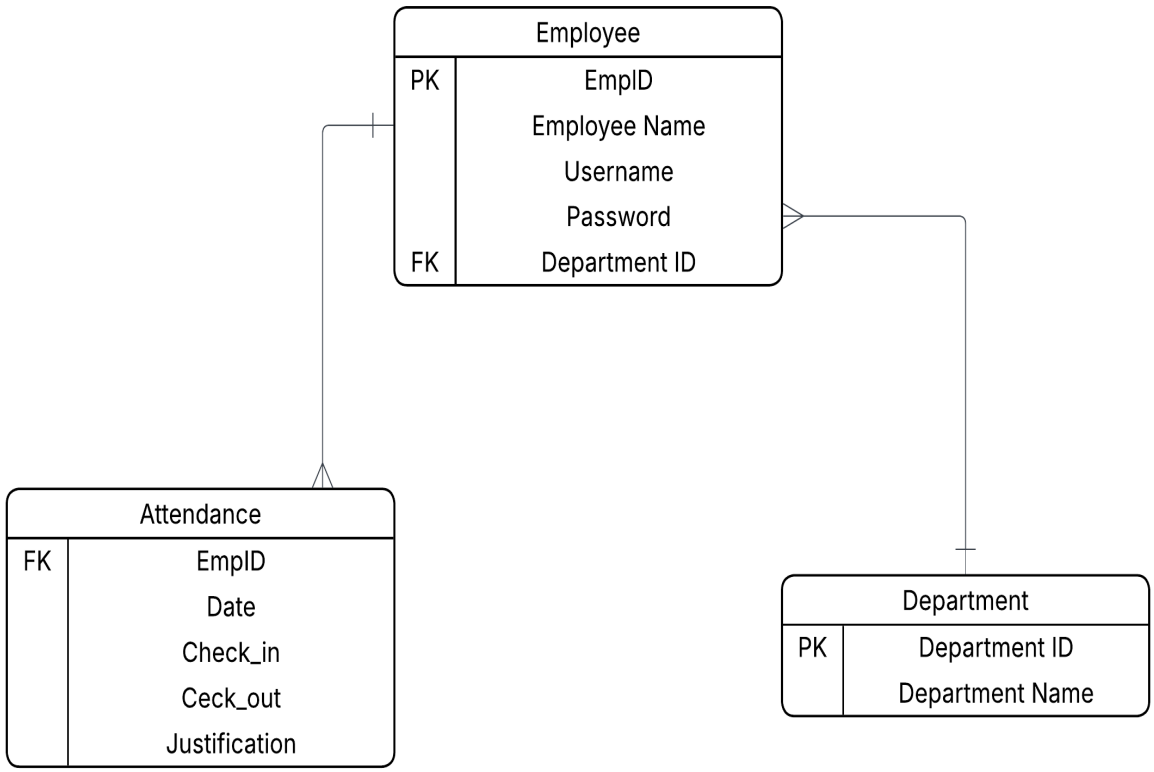


Figure 3.1 Database Diagram

4. Business Workflow Process

The Business Process Workflow represents the sequence of operations performed within the *Employee Attendance Management System,* It illustrates how users (employees and administrators) interact with the system from login and attendance check-in to report generation and management actions. This ensures a clear understanding of system logic and user roles

4.1 Employee Workflow

1. **Login to System:** The employee enters valid credentials (Username and Password) If authenticated, the system grants access to the employee dashboard.
2. **Check-in:** The employee records the start of their working day by clicking the Check-in button, The system automatically logs the current time and date in the Attendance table.
3. **Check-out:** At the end of the day, the employee clicks the Check-out button. The system records the check-out time and calculates total worked hours.
4. **View Dashboard:** The employee can view a summary of attendance data (on-time, late, early check-out) through the performance dashboard.
5. **Add Justification (if flagged):** If a record is marked as late or early, the employee may submit a justification note for review.

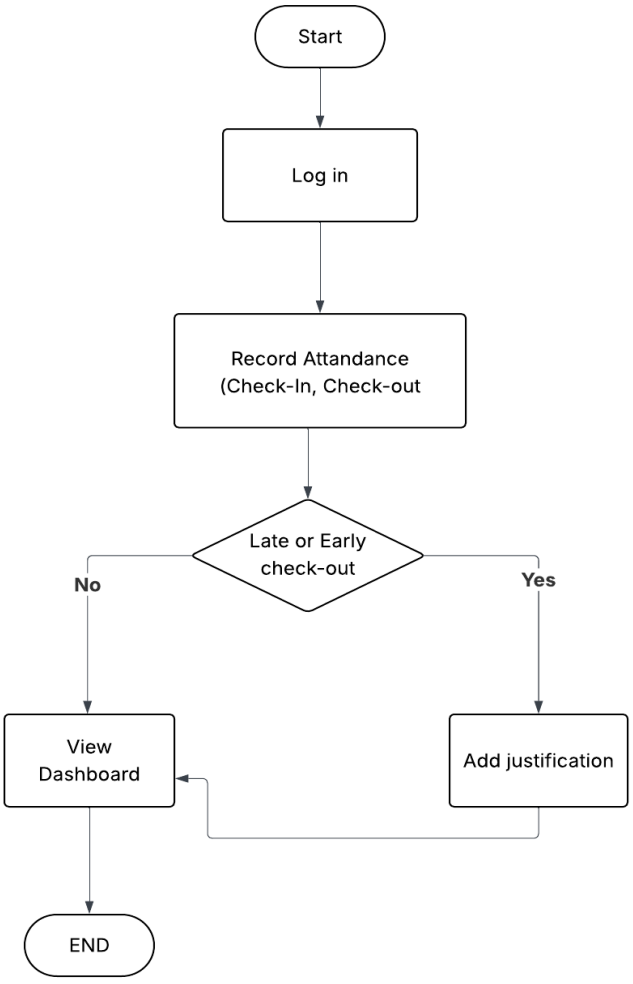


Figure 4.1 Workflow

4.2 Admin Workflow

1. **Login to System:** Admin logs in using authorized credentials
2. **Manage Employees:** Admin can add, update, or delete employee information.
3. **Manage Departments:** Admin can create, edit, or remove departments as required.
4. **Monitor Attendance:** Admin can view all employees' attendance records and check performance statistics per department.

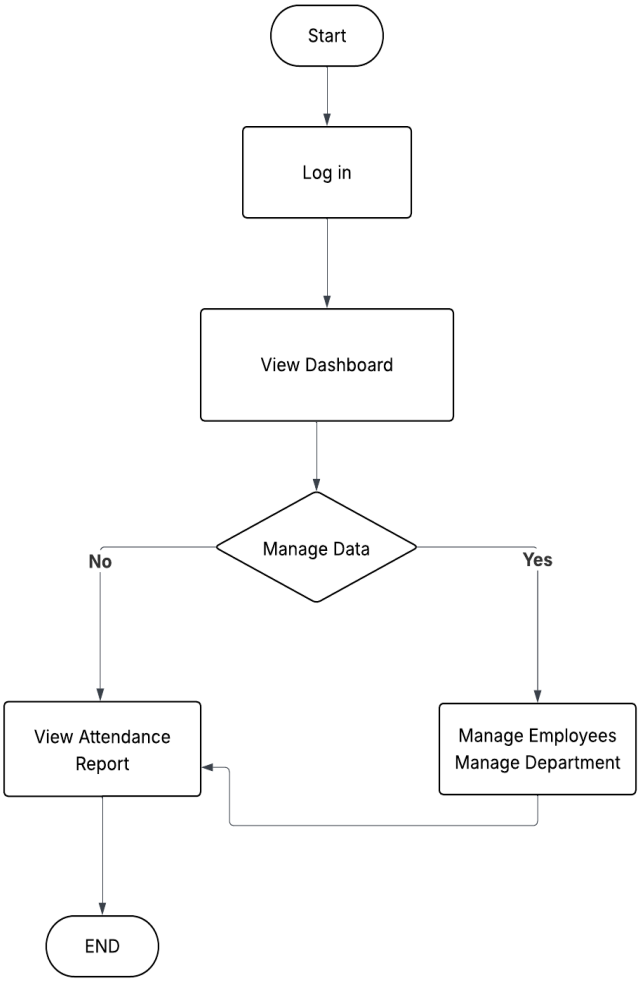


Figure 4.2 Workflow

4.3 Process Flow Description

The workflow follows a top-down logical structure, starting from user authentication and branching into two main roles:

Employee Role: focuses on attendance recording and personal performance.

Admin Role: focuses on system management and analysis.

Each step updates the database automatically (Employee, Department, and Attendance tables), ensuring data consistency and traceability

5. User Acceptance test

The Blackbox Testing method was applied to verify the system's functionality from the user's perspective. This test focused on ensuring that each feature of the Employee Attendance Management System such as login, attendance tracking, and management operations works correctly without checking the internal code structure. The following table summarizes the main UAT scenarios and their results:

Test case ID	Related FR	Scenario	Test Step	Expected Result	Status
TC 1	**FR-2**	Employee Login	Enter valid username and password. 2. Click "Login	system authenticates credentials and redirects employee to dashboard	PASS
TC 2	**FR-2**	Invalid Login Attempt	enter incorrect username or password. 2. Click "Login.	system displays error message "Invalid username or password.	PASS
TC 3	**FR-2**	Check-In Attendance	Employee clicks "Check-In	system records check-in time and date successfully	PASS
TC 4	**FR-2**	Check-Out Attendance	Employee clicks "Check-Out	system records check-out time and calculates total worked hours	PASS
TC 5	**FR-3**	Late Check-In Detection	Employee performs check-in after 8:30 AM	System flags the "record as "Late	PASS
TC 6	**FR-3**	Early Check-Out Detection	Employee performs check-out before 3:00 PM	system flags the record as "Early Check-out	PASS
TC 7	**FR-2**	submit Justification	Employee Submit justification for flagged attendance. 2. Click "Save	system stores justification and marks record as "Justified.	PASS
TC 8	**FR-1**	Admin – Add New Employee	1. Admin navigates to Employee Page. 2. Click "Add Employee." 3. Enter details and s	new employee appears in employee list	PASS
TC 9	**FR-1**	Admin - manage Employees	1.Admin navigates to Employees Page. 2.Click "Manage" 3. Add/Edit/Delete Employees.	System updates the employees records successfully	PASS
TC 10	**FR-1**	Admin - manage Departments	1.Admin navigates to Departments Page. 2.Click "Manage" 3. Add/Edit/Delete Departments.	System updates the Departments records successfully	PASS
TC 11	**FR-1**	Admin – View Attendance Dashboard	Admin opens dashboard view	Dashboard displays accurate attendance and department statistics	PASS
TC 12	**FR-3**	Weekly working hours Calculation	Employee completes daily attendance for 5 working days	System calculates total weekly hours (target = 40 hrs.)	PASS

Table 5.1 User Acceptance test